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INDRAPRASTHA INSTITUTE OF INFORMATION TECHNOLOGY DELHI
(A State University Established by Govt. of NCT Delhi)

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पाठ्यक्रम सं.
Course No. C S E 6 3 8

ग्रुप संख्या
Group No.

पाठ्यक्रम
Course Title Graduate Systems

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प्रयोग किए गए अनुवर्ती पृष्ठों की संख्या
No. of continuation sheets used

प्रश्न सं. Q. No.	प्राप्त अंक Marks
1.	4
2.	2
3.	NA
4.	4
5.	3
6.	0
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HONOR CODE

I pledge to follow the Honor Code and to obey all rules for writing examinations as specified in the Institute guidelines and as specified by the course Instructor. I understand that when asked to follow the Honor Code on examinations, it implies that

1. **I will work entirely alone on examination.**
2. **I will not carry and use any electronic gadget (mobile phones, watch etc.) during the exams.**
3. **I will no carry any cheating material inside the examination hall.**
4. **It is my responsibility to obtain clarification from the Instructor if there are questions concerning the requirements of the Honor Code.**

Student Signature (With Date) VSRaju 24/2/26.

सभी पृष्ठों पर लिखें / **Write on all pages**

पाठ्यक्रम निर्धारक के हस्ताक्षर और दिनांक
Signature of Course co-ordinator and date

Q1.

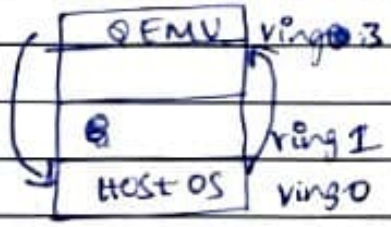
(a) CPU generated ✗

(b) A user program which is written in a high level language like C, C++, Java ... are initially in userSpace and executing low privilege task with help of CPU so to invoke a Kernel code which is a high privilege task can be done with the help of System Calls (a software trap created for accessing system resources like Memory, I/o and Kernel code). So with the system call, which running executable code, OS looks at the IDT (Interrupt Descriptor table) which tells where exactly the Kernel code is in the RAM and the execution happens and process will still be considered to be in running state. (+2)

(c) User programs obtains the address of Kernel code from the IDT (Interrupt Descriptor table) where each Interrupt is mapped to a unique number and that number is mapped to starting address of the Kernel code of that particular interrupt which is physical address.
We will not make user to run his own code by using virtual addressing because user may be malicious and accessing Main memory directly is problematic, So, IDT is used intermediately to serve the interrupt. This is called ISR (Interrupt Service Routine). (+2)

d).

Q2. Running QEMU+KVM is faster than running QEMU in full emulation mode because, In emulation.



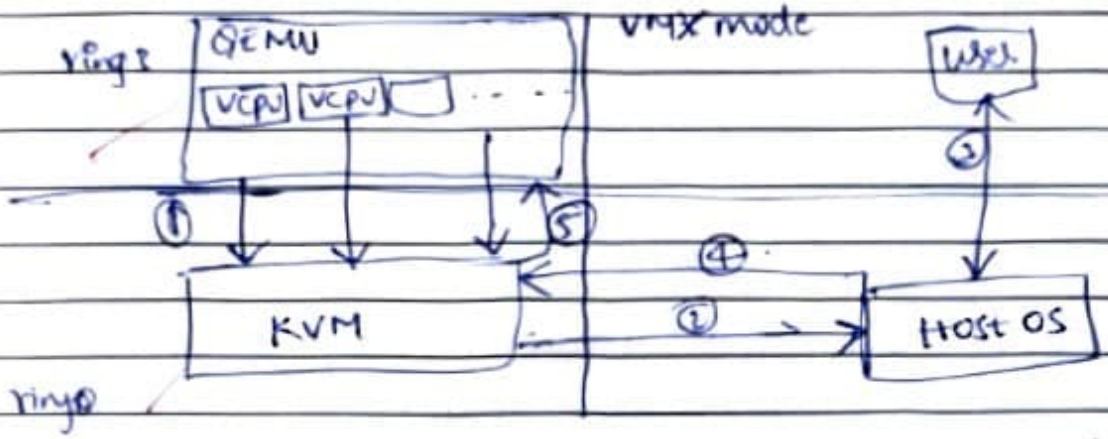
here QEMU is a user process, which means QEMU as a user program it can only execute low priviledge task and for executing any high priviledge tasks, it should get trap emulated to lower ring levels i.e to Host OS inorder to finish that task.

which is a too much of context switching and takes lots of time here

So, If we use a VMM like KVM, It provides KVM-run, KVM-launch, KVM-close such type of techniques ~~do~~ ~~here~~ which make it easier and fast to communicate with host OS.

Here QEMU/KVM, QEMU is in user mode /ring 3 KVM is in ring0, when a certain high priviledged task need to be done QEMU sends ioctl signal to KVM

KVM switches to VMX mode where HOST OS is communicated and it helps in performing high privilege ops



Here, 3 types of connections are established between QEMU and KVM, VCPU's are like threads which makes computation faster and parallel.

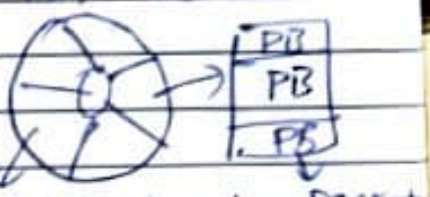
Not right question exactly

Q4. Once packet is received at the NIC's buffer i.e the hardware buffer then:

① NIC checks the matches the MAC address with a set of MAC address it has and performs Check Sum. If both valid then proceed.

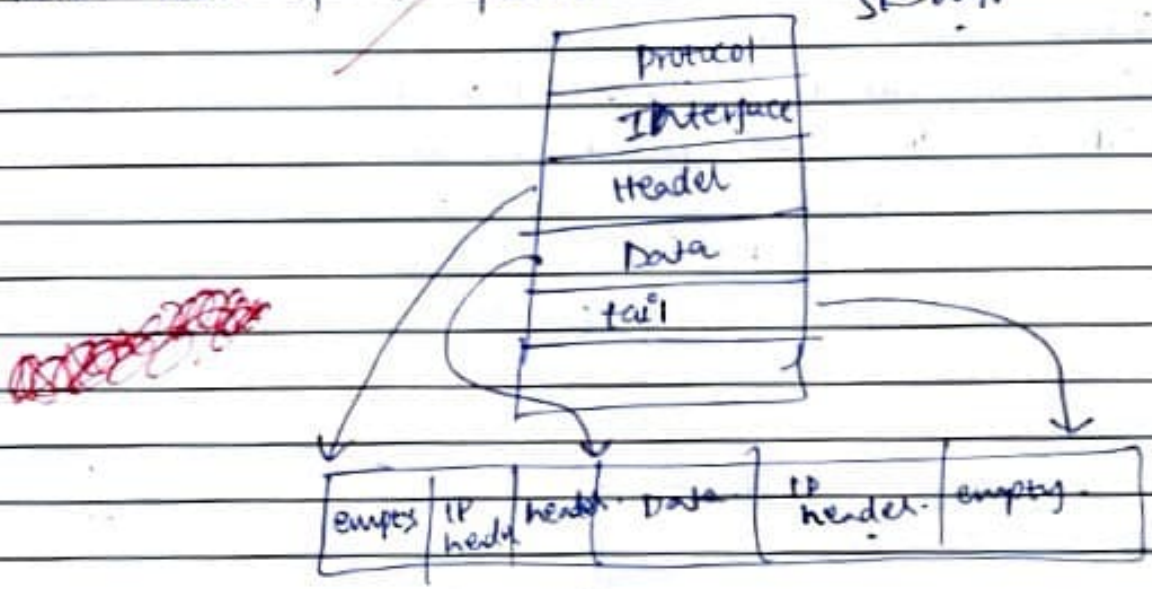
② Now, we are at L2 (Data link layer), here DMA (direct memory Access) helps in placing the packets on to the packet buffer array.

③ Here as we have two queues Tx, Rx, we use Tx now, which is a circular queue. This points the ~~packet buffer~~ array. and



packet length + Packet data. Buffer

④ Now a special Data Structure called SK-buffer (Skb) is created to store the meta data of a packet.



this will get initialized for each packet and enqueued to the queue which we use to send data to application.

⑤ L3, IP.
→ checks IP ~~for~~ ~~socket~~ socket.
→ Removes them (headers)
→ Fragment the data if needed.

and now transmission TCP protocol will check state machine and enqueue it.

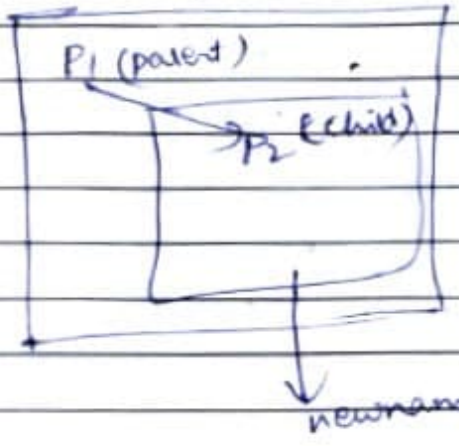
Now the encrypted data packets are sent to destination and the SK_buff is released.

(4)

(5). (a) I will use clone() because rather than fork() because when using fork() if we create, a copy of parent memory image is given to child but the memory is not shared after copy on write.

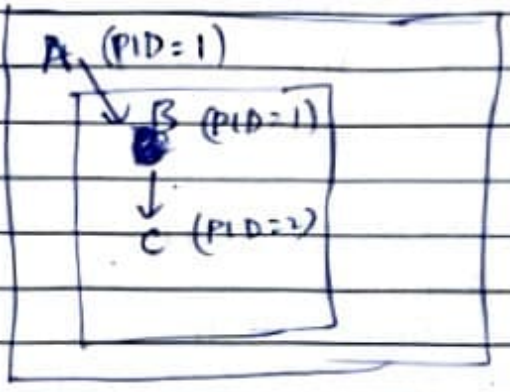
(2)

while on clone() the child is placed into a new namespace and the parent is still pointing it



(b).

(c)



(i) PID of B is 1 because that is first precess in the newly created namespace 1

(ii) "C" will be reaped by "A" as A is the parent namespace of the PID 1 process universal namespace.

(6) MAC address is unique to a machine, which is binded to the NIC on the machine which is used to check the MAC address of data packet when received and encapsulated when data is send and is taken care by NIC in L_1.

where IP is private and public IP. to communicate globally. private IP is given by : Internet Service provider and is unique within a network. mostly in 192.168.0.0 or 10.0.0.0. while public IP is assigned by NAT when a IP from a network is

trying to communicate with global servers out that network. this ensure less usage of IPs and security guaranteed.

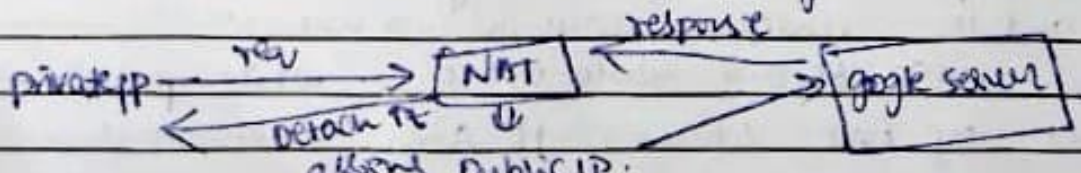
(b) port address is dedicated to each applications on one machine which is responsible for sending and listening to the results from different IP.

a (IP + port) is called a socket, which is uniquely identifiable by NAT and can be properly communicated.

(SrcIP, Source source port, Destination IP, Destination port) 4-tuple is a unique ~~trans~~ network transmission in the Internet.

(c) yes, IP addresses are IPv4 addressed and IPv6 addressed where IPv4 are 32 bit address given by a service provided and are only 2^{32}, which are not sufficient for recent future, so shift to IPv6 is taking place where it is 2^{128} IPs. (128 bits).

Also, private IP and public IP.
here private IP is within an organisation and is unique within an organisation mostly in 192.168.0.0 & 10.0.0.0 where as public IP is known to all over internet and used to communicate with globle servers like google.



This ensures limited use of IPs and security intra network